12.11 Stamps, Cups and e

WORKSHEET

The question

I have a Penny Black stamp which I want to sell to my friend Jim, but we cannot agree a price. So I put the stamp under one of two cups, jumble them up, and let Jim guess which one it is under. If he guesses correctly, I add a third cup, jumble them up, and let Jim guess correctly, adding another cup each time. The price he pays for the stamp is £N, where N is the number of cups present when Jim fails to guess correctly. Find $\mathrm{P}(N = k)$. Show that $\mathrm{E}(N) = e$ and calculate $\mathrm{Var}(N)$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself For $N = 4$, what must happen? With what probability?

STAGE 2 The expectation

Ask yourself Simplify $k \cdot \frac{k-1}{k!}$. What series do you get?

STAGE 3 The variance

Ask yourself Find $\mathrm{E}(N^2)$ by writing $k^2 = k(k-1) + k$, or by simplifying $\frac{k^2(k-1)}{k!}$ directly.

Try this next

1. **The generating function.** Show that $\mathrm{E}(t^N) = 1 + (t-1)e^t$, and use it to find $\mathrm{E}(N)$ and $\mathrm{E}(N(N-1))$.
2. **The tail formula.** Prove that $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k)$ for a random variable taking non-negative integer values, and use it to find $\mathrm{E}(N)$ in one line.
3. **Starting with one cup.** If the game starts with one cup (which Jim cannot get wrong), how does the distribution change? What is the expected price?
4. **The simplex.** Show that the volume of $\{u_1 + u_2 + u_3 < 1, u_i \geq 0\}$ is $\frac{1}{6}$, and use induction to show the k-dimensional version is $\frac{1}{k!}$.

Further reading

- **Eli Maor, *e: The Story of a Number*** (Princeton University Press). The history of e, including its appearances in probability.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on generating functions.
- **Paul Nahin, *Digital Dice*** (Princeton University Press). Probability puzzles solved by Monte Carlo simulation.

The full guided solution, and what the question leads to at university, is in Section 12.11:
https://beyondstep.keithmcnulty.org/ch12/stamps-and-e.html